Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement : Patient Vitals Alert & Monitoring App

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1. Architectural Style Evaluation:

The three candidate architectural styles — Layered, Microservices, and Client-Server — are compared below against the specific functional and non-functional requirements of the Patient Vitals Alert & Monitoring App: continuous ingestion of up to 500 concurrent telemetry streams (NFR-001), threshold breaches flagged within 2 seconds (FR-001), guaranteed 60-second SMS escalation (FR-003), and HIPAA-grade TLS/AES-256 protection of patient data (NFR-002).

Criteria	Layered Architecture	Microservices Architecture	Client-Server Architecture
Structure	Presentation, Business (Threshold/Alert logic), and Data layers stacked in one deployable application.	Independently deployable services: Ingestion, Evaluation, Escalation, Configuration, Acknowledgment, Dashboard, each with its own data access.	A central vitals server handles all logic; caregiver and patient apps act as thin clients connecting to it.
Pros for this scenario	Simple to build and reason about; adequate for a small, single-team prototype.	Ingestion and alerting can scale to 500+ concurrent streams (NFR-001) independently of the dashboard/config services; a failure in one service does not stop critical alerting; each service can enforce its own TLS/AES-256 boundary (NFR-002).	Simple to deploy and keep data consistent since there is one authoritative server.
Cons for this scenario	The whole application must be scaled and redeployed together, so a dashboard traffic spike can slow down the 2-second alert pipeline (FR-001); one failure can take the entire monitoring app down — unacceptable for a life-critical alert path.	More operational complexity (multiple deployments, service discovery) and inter-service network latency to manage.	The single server is a single point of failure and a scaling bottleneck — directly conflicts with NFR-001's 500-stream, 99.99%-uptime requirement, and one overloaded caregiver session could delay a critical alert to another patient.
Fit for Patient Vitals Alert	Poor — safety-critical alerting (FR-001, FR-003)	Best fit — matches the	Poor — a single point of

& Monitoring App	cannot risk being coupled to and slowed by less critical features.	scaling, fault-isolation, and independent-security-boundary needs of a real-time, life-critical, HIPAA-regulated system	failure is not acceptable when missed alerts can endanger patient safety.

2. Architecture Selection & Justification

Architecture Selection: *We chose **Microservices Architecture** for the Patient Vitals Alert & Monitoring System.*

Reason 1 — Independent Scaling for Real-Time Ingestion

NFR-001 requires the telemetry gateway to sustain at least 500 concurrent continuous vitals streams at 99.99% uptime, while FR-001 requires threshold breaches to be flagged within 2 seconds. Microservices let the Telemetry Ingestion Service and Vitals Evaluation Engine be scaled independently of lower-traffic services such as Threshold Configuration or the Dashboard, so ingestion throughput can grow with the patient population without over-provisioning the entire system.

Reason 2 — Fault Isolation for Life-Critical Alerting

FR-003 requires that an unacknowledged alert automatically escalates via SMS within 60 seconds. Because each capability (ingestion, evaluation, escalation, acknowledgment, dashboard, configuration) runs as an independently deployable service, a failure or slowdown in a non-critical service, such as the Caregiver Dashboard, cannot block or delay the Alert & Escalation Service. This isolates the safety-critical escalation path from the rest of the system — a guarantee that a monolithic layered design or a single client-server instance cannot provide.

Security Implication

Splitting the system into narrowly scoped services allows each service boundary to enforce TLS 1.2+ in transit and AES-256 at rest independently (NFR-002), and to expose only the minimum interface it needs — for example, the Threshold Configuration Service exposes only ThresholdConfig and never gets direct database access. This reduces the blast radius of a compromised service and keeps HIPAA-regulated patient data access auditable at each service boundary rather than across one large shared codebase.

Performance Implication

Because the Vitals Evaluation Engine and Alert & Escalation Service are deployed and scaled independently of the Dashboard and Configuration services, ingestion-and-alerting latency stays low and predictable — meeting the 2-second flagging target (FR-001) and the 60-second escalation target (FR-003) — even during traffic spikes on lower-priority services, since compute resources are provisioned per-service rather than shared.

Trade-off Acknowledged

Microservices introduce operational complexity and network latency between services. This is accepted because the alternative (Layered or Client-Server) would couple the safety-critical alerting path to lower-priority features and create a single scaling/failure bottleneck, which is unacceptable for a life-critical monitoring system.